7.0 Switching

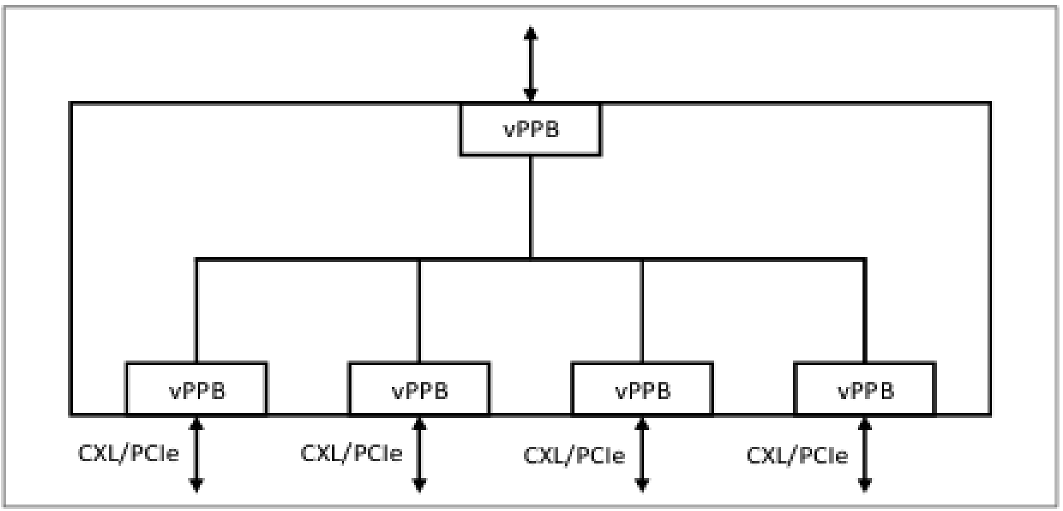
7.1 Overview

This section provides an architecture overview of different CXL switch configurations.

7.1.1 Single VCS

A single VCS consists of a single CXL Upstream Port and one or more Downstream Ports as illustrated in Figure 7-1.

Figure 7-1. Example of a Single VCS



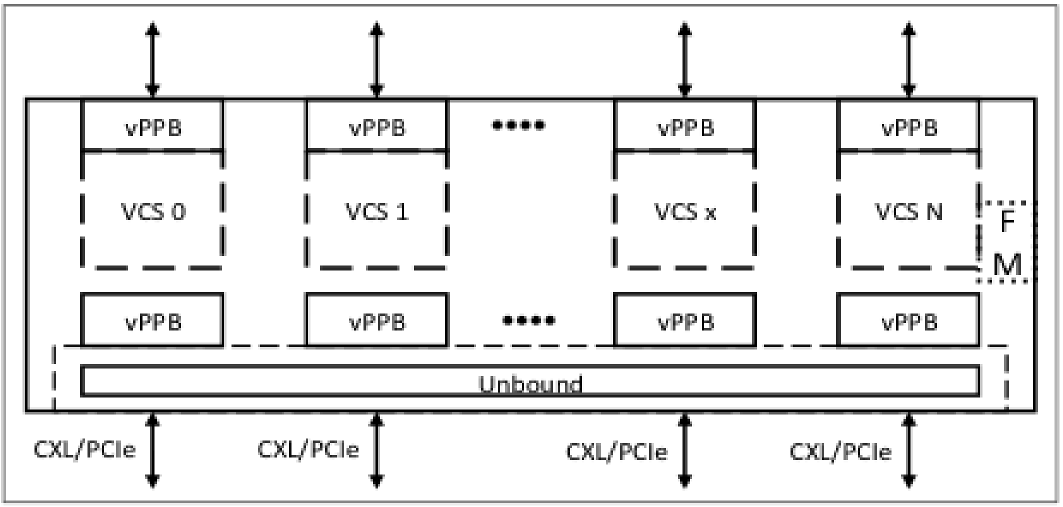
A Single VCS is governed by the following rules:

- Must have a single USP
- Must have one or more DSPs
- DSPs must support operating in CXL mode or PCIe* mode
- All non-MLD (includes PCIe and SLD) ports support a single Virtual Hierarchy below the vPPB
- Downstream Switch Port must be capable of supporting RCD mode
- Must support the CXL Extensions DVSEC for Ports (see Section 8.1.5)
- The DVSEC defines registers to support CXL.io decode to support RCD below the Switch and registers for CXL Memory Decode. The address decode for CXL.io is in addition to the address decode mechanism supported by vPPB.
- Fabric Manager (FM) is optional for a Single VCS

7.1.2 Multiple VCS

A Multiple VCS consists of multiple Upstream Ports and one or more Downstream Ports per VCS as illustrated in Figure 7-2.

Figure 7-2. Example of a Multiple VCS with SLD Ports



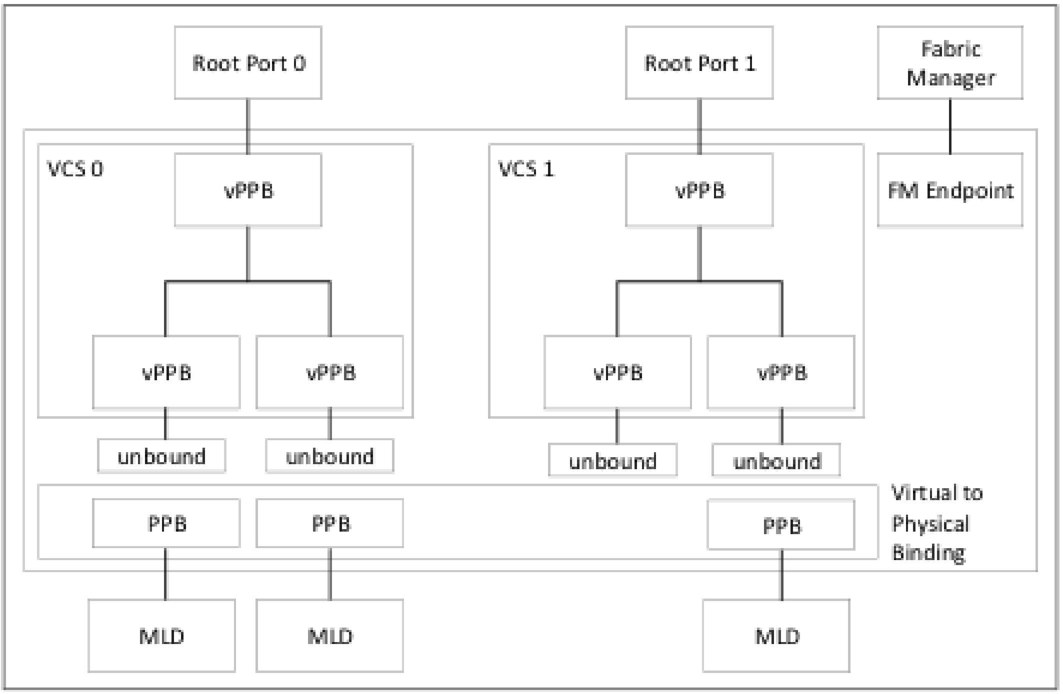
A Multiple VCS is governed by the following rules:

- Must have more than one USP.
- Must have one or more DS vPPBs per VCS.
- The initial binding of upstream (US) vPPB to physical port and the structure of the VCS (including number of vPPBs, the default vPPB capability structures, and any initial bindings of downstream (DS) vPPBs to physical ports) is defined using switch vendor specific methods.
- Each DSP must be bound to a PPB or vPPB.
- FM is optional for Multiple VCS. An FM is required for a Multiple VCS that requires bind/unbind, or that supports MLD ports. Each DSP can be reassigned to a different VCS through the managed Hot-Plug flow orchestrated by the FM.
- When configured, each USP and its associated DS vPPBs form a Single VCS Switch and operate as per the Single VCS rules.
- DSPs must support operating in CXL mode or PCIe mode.
- All non-MLD, non-Fabric, and non-GFD HBR ports support a single Virtual Hierarchy below the Downstream Switch Port.
- DSPs must be capable of supporting RCD mode.

7.1.3 Multiple VCS with MLD Ports

A Multiple VCS with MLD Ports consists of multiple Upstream Ports and a combination of one or more Downstream MLD Ports, as illustrated in Figure 7-3.

Figure 7-3. Example of a Multiple Root Switch Port with Pooled Memory Devices



A Multiple VCS with MLD Ports is governed by the following rules:

- More than one USP.
- One or more Downstream vPPBs per VCS.
- Each SLD DSP can be bound to a Single VCS.
- An MLD-capable DSP can be connected to up to 16 USPs.
- Each of the SLD DSPs can be reassigned to a different VCS through the managed Hot-Plug flow orchestrated by the FM.
- Each of the LD instances in an MLD component can be reassigned to a different VCS through the managed Hot-Plug flow orchestrated by the FM.
- When configured, each USP and its associated vPPBs form a Single VCS, and operate as per the Single VCS rules.
- DSPs must support operating in CXL mode or PCIe mode.
- All non-MLD ports support a single Virtual Hierarchy below the DSP.
- DSPs must be capable of supporting RCD mode.

7.1.4 vPPB Ordering

vPPBs within a VCS are ordered in the following sequence: the USP vPPB, then the DSP vPPBs in increasing Device Number, Function Number order. This means Function 0 of all vPPBs in order of Device Number, then all vPPBs enumerated at Function 1 in order of Device Number, etc.

For a switch with 65 DSP vPPBs whose USP vPPB was assigned a Bus Number of 3, that would result in the following vPPB ordering:

vPPB #	PCIe ID
0	USP 3:0.0
1	DSP 4:0.0
2	DSP 4:1.0
3	DSP 4:2.0
	...
32	DSP 4:31.0
33	DSP 4:0.1
34	DSP 4:1.1
	...
64	DSP 4:31.1
65	DSP 4:0.2

This ordering also applies in cases where multi-function vPPBs exist but not all 32 Device Numbers are assigned. For example, a switch with 8 DSP vPPBs whose USP vPPB was assigned a Bus Number of 3 could present its DSP vPPBs in such a way that the host enumeration would result in the following vPPB ordering:

vPPB #	PCIe ID
0	USP 3:0.0
1	DSP 4:0.0
2	DSP 4:1.0
3	DSP 4:2.0
4	DSP 4:0.1
5	DSP 4:1.1
6	DSP 4:2.1
7	DSP 4:0.2
8	DSP 4:1.2

7.2 Switch Configuration and Composition

This section describes the CXL switch initialization options and related configuration and composition procedures.

7.2.1 CXL Switch Initialization Options

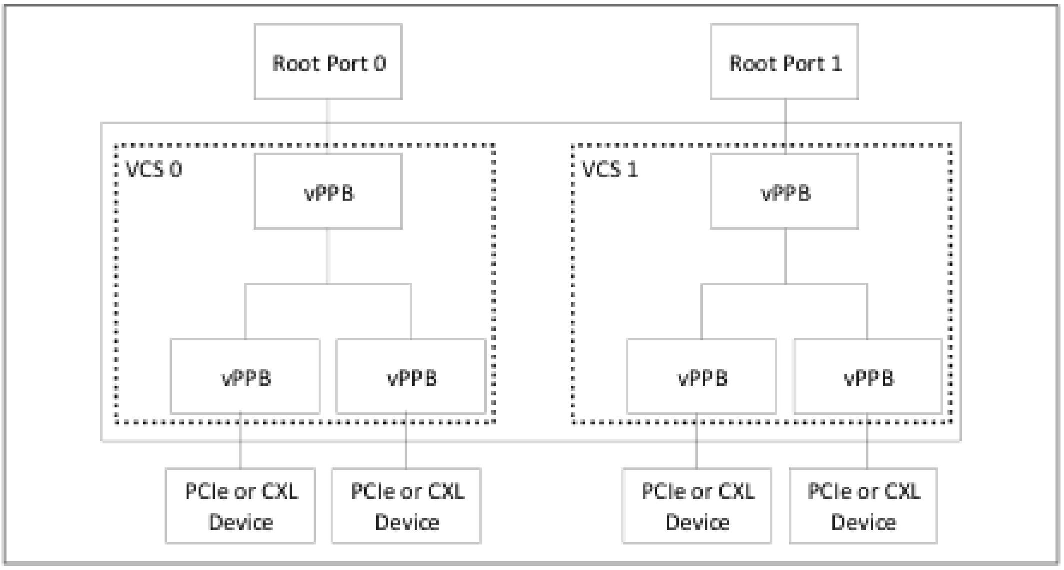
The CXL switch can be initialized using three different methods:

- Static
- FM boots before the host(s)
- FM and host boot simultaneously

7.2.1.1 Static Initialization

Figure 7-4 shows a statically initialized CXL switch with 2 VCSs. In this example, the downstream vPPBs are statically bound to ports and are available to the host at boot. Managed hot-add of Devices is supported using standard PCIe mechanisms.

Figure 7-4. Static CXL Switch with Two VCSs



Static Switch Characteristics:

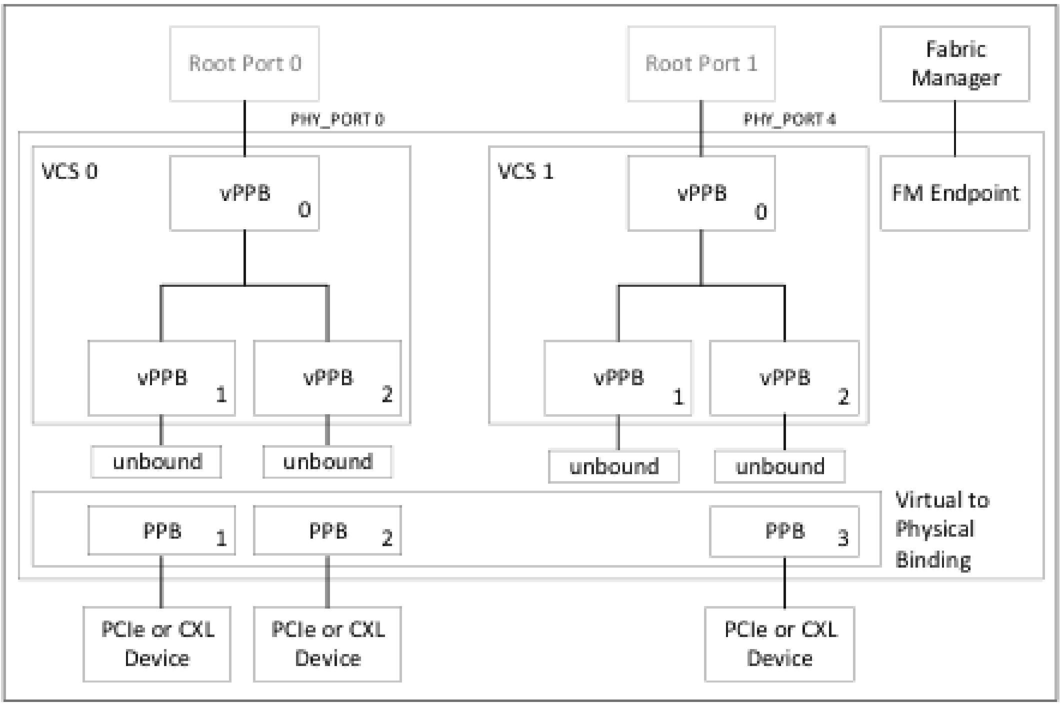
- No support for MLD Ports
- No support for rebinding of ports to a different VCS
- No FM is required
- At switch boot, all VCSs and Downstream Port bindings are statically configured using switch vendor defined mechanisms (e.g., configuration file in SPI Flash)
- Supports RCD mode, CXL VH mode, or PCIe mode
- VCSs, including vPPBs, behave identically to a PCIe switch, along with the addition of supporting CXL protocols
- Each VCS is ready for enumeration when the host boots
- Hot-add and managed hot-remove are supported
- No explicit support for Async removal of CXL devices; Async removal requires that root ports implement CXL Isolation (see Section 12.3)

A switch that provides internal Endpoint functions is beyond the scope of this specification.

7.2.1.2 Fabric Manager Boots First

In cases where the FM boots first (prior to host(s)), the switch is permitted to be initialized as described in the example shown in Figure 7-5.

Figure 7-5. Example of CXL Switch Initialization when FM Boots First



1. FM boots while hosts are held in reset.
2. All attached DSPs link up and are bound to FM-owned PPBs.
3. DSPs link up and the switch notifies the FM using a managed hot-add notification.

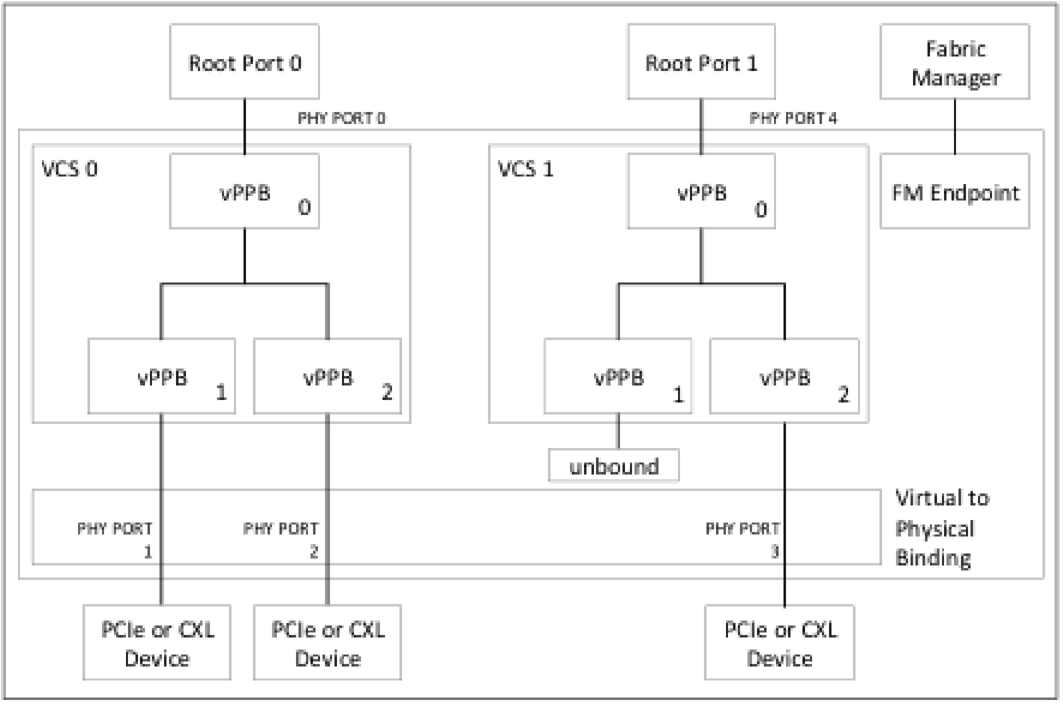


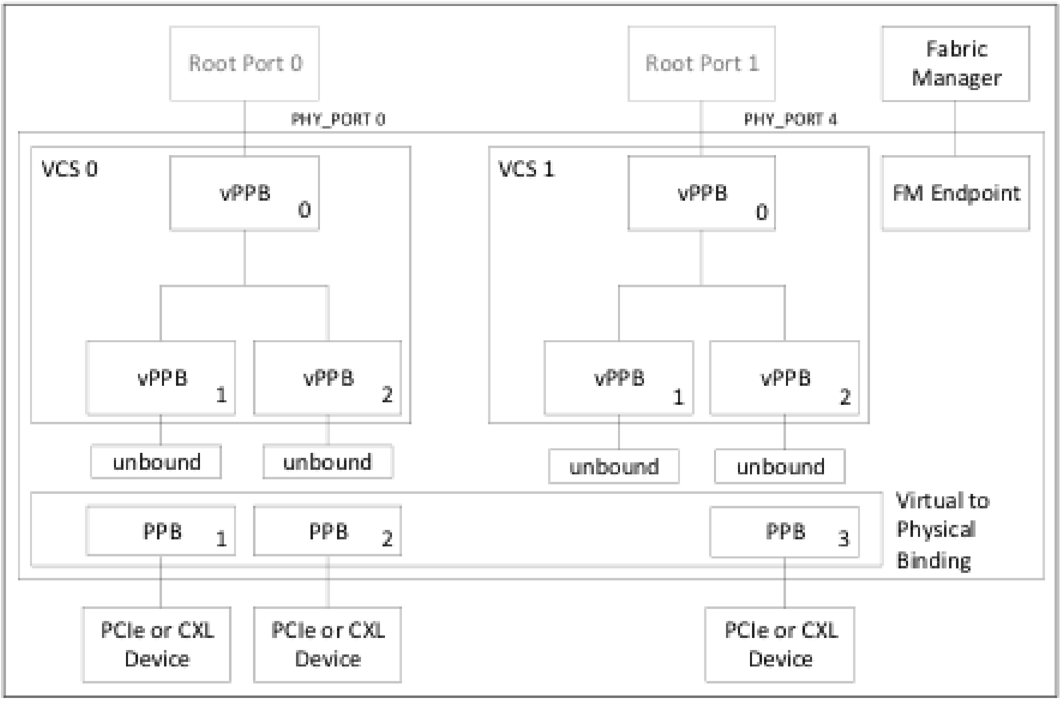
Figure 7-6. Example of CXL Switch after Initialization Completes

As shown in the example above in Figure 7-6, the following steps are taken to configure the switch after initialization completes:

1. FM sends bind command BIND (VCS0, vPPB1, PHY_PORT_ID1) to the switch. The switch then configures virtual to physical binding.
2. Switch remaps vPPB virtual port numbers to physical port numbers.
   — Virtual port number is the index of the vPPB (as specified in the Bind vPPB command discussed in Section 7.6.7.2.2) per virtual hierarchy.
3. Switch remaps vPPB connector definition (PERST#, PRSNT#) to physical connector.
4. Switch disables the link using PPB Link Disable.
5. At this point, all Physical downstream PPB functionality (e.g., Capabilities, etc.) maps directly to the vPPB including Link Disable, which releases the port for linkup.
6. The FM-owned PPB no longer exists for this port.
7. When the hosts boot, the switch is ready for enumeration.

7.2.1.3 Fabric Manager and Host Boot Simultaneously

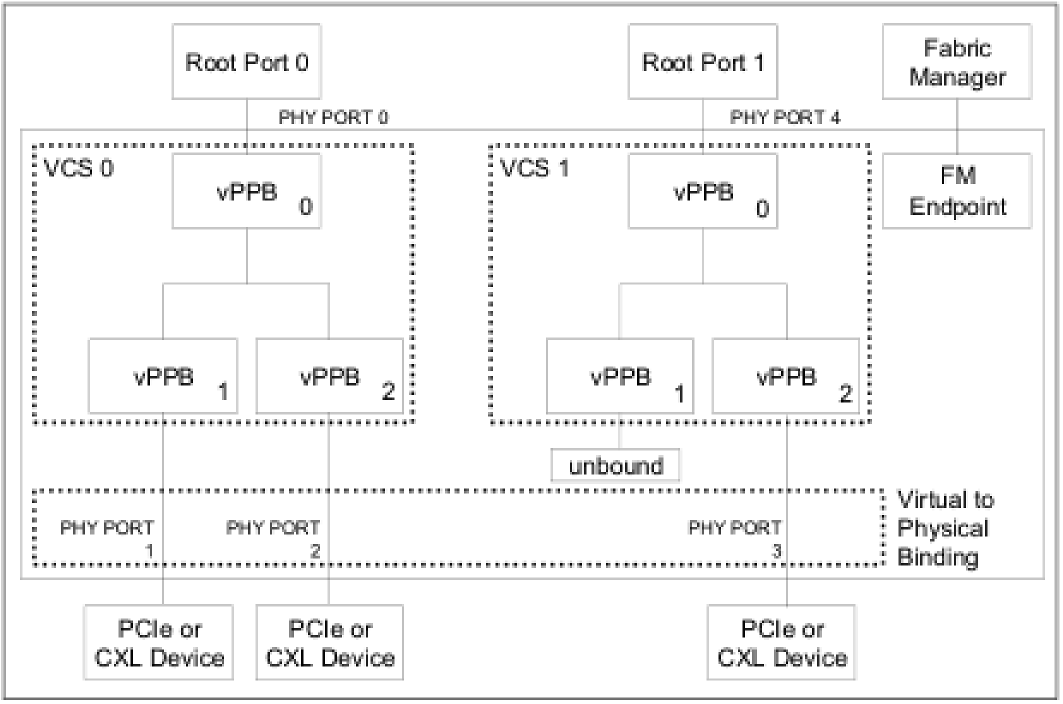
Figure 7-7. Example of Switch with Fabric Manager and Host Boot Simultaneously



In the case where the switch, FM, and host boot at the same time:

1. VCSs are statically defined.
2. DSP vPPBs within each VCS are unbound and presented to the host as Link Down.
3. Switch discovers downstream devices and presents them to the FM.
4. Host enumerates the VH and configures the DVSEC registers.
5. FM performs port binding to vPPBs.
6. Switch performs virtual to physical binding.
7. Each bound port results in a hot-add indication to the host.

Figure 7-8. Example of Simultaneous Boot after Binding



7.2.2 Sideband Signal Operation

The availability of slot sideband control signals is decided by the form-factor specifications. Any form factor can be supported, but if the form factor supports the signals listed in Table 7-1, the signals must be driven by the switch or connected to the switch for correct operation.

All other sideband signals have no constraints and are supported exactly as in PCIe.

Table 7-1. CXL Switch Sideband Signal Requirements

Signal Name	Signal Description	Requirement
USP PERST#	PCIe Reset provides a fundamental reset to the VCS	This signal must be connected to the switch if implemented
USP ATTN#	Attention button indicates a request to the host for a managed hot-remove of the switch	If hot-remove of the switch is supported, this signal must be generated by the switch
DSP PERST#	PCIe Reset provides a power-on reset to the downstream link partner	This signal must be generated by the switch if implemented
DSP PRSNT#	Out-of-band Presence Detect indicates that a device has been connected to the slot	This signal must be connected to the switch if implemented
DSP ATTN#	Attention button indicates a request to the host for a managed hot-remove of the downstream slot	If managed hot-remove is supported, this signal must be connected to the switch

This list provides the minimum sideband signal set to support managed Hot-Plug. Other optional sidebands signals such as Attention LED, Power LED, Manual Retention Latch, Electromechanical Lock, etc. may also be used for managed Hot-Plug. The behavior of these sideband signals is identical to PCIe.

7.2.3 Binding and Unbinding

This section describes the details of Binding and Unbinding of CXL devices to a vPPB.

7.2.3.1 Binding and Unbinding of a Single Logical Device Port

A Single Logical Device (SLD) port refers to a port that is bound to only one VCS. That port can be linked up with a PCIe device or a CXL Type 1, Type 2, or Type 3 SLD component. In general, the vPPB bound to the SLD port behaves the same as a PPB in a PCIe switch. An exception is that a vPPB can be unbound from any physical port. In this case the vPPB appears to the host as if it is in a Link Down state with no Presence Detect indication. If optional rebinding is desired, this switch must have an FM API support and FM connection. The Fabric Manager can bind any unused physical port to the unbound vPPB. After binding, all the vPPB port settings are applied to that physical port.

Figure 7-9 shows a switch with bound DSPs.

Figure 7-9. Example of Binding and Unbinding of an SLD Port

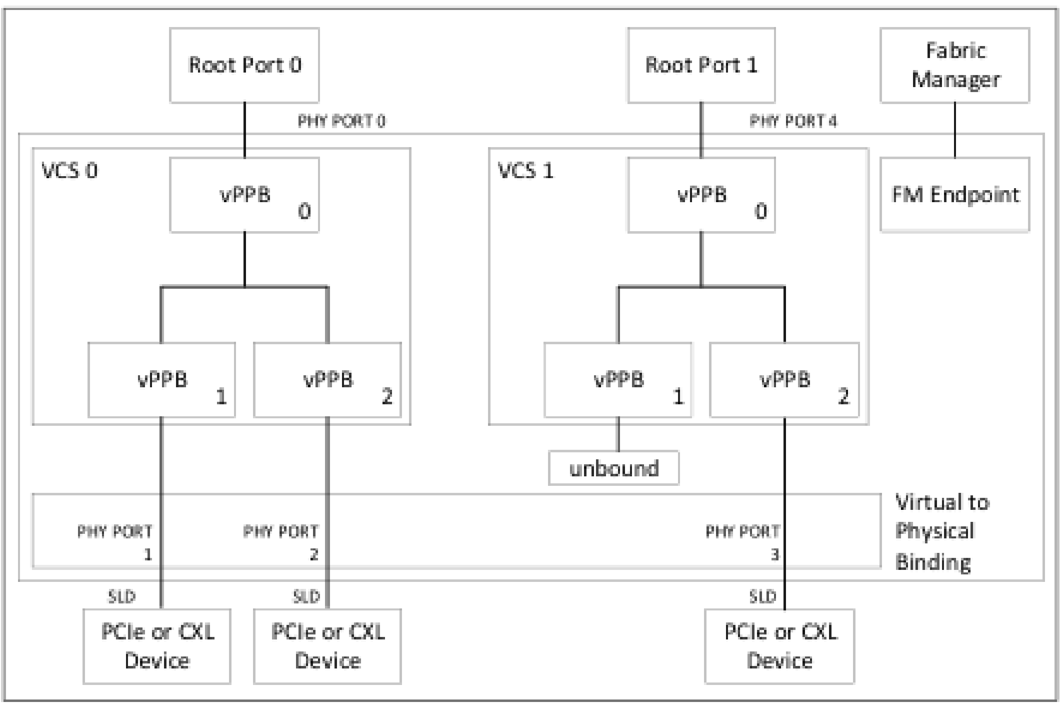


Figure 7-10 shows the state of the switch after the FM has executed an unbind command to vPPB2 in VCS0. Unbind of the vPPB causes the switch to assert Link Disable to the port. The port then becomes FM-owned and is controlled by the PPB settings for that physical port. Through the FM API, the FM has CXL.io access to each FM-owned SLD port or FM-owned LD within an MLD component. The FM can choose to prepare the logical device for rebinding by triggering FLR or CXL Reset. The switch prohibits any CXL.io access from the FM to a bound SLD port and any CXL.io access from the FM to a bound LD within an MLD component. The FM API does not support FM generation of CXL.cache or CXL.mem transactions to any port.

Figure 7-10. Example of CXL Switch Configuration after an Unbind Command

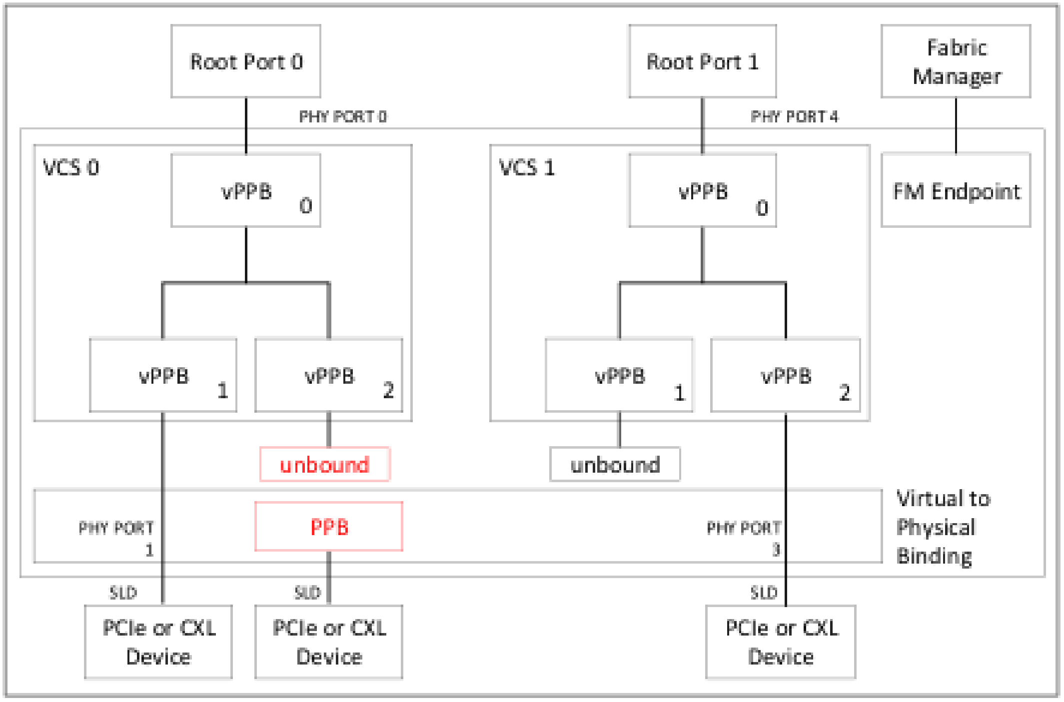
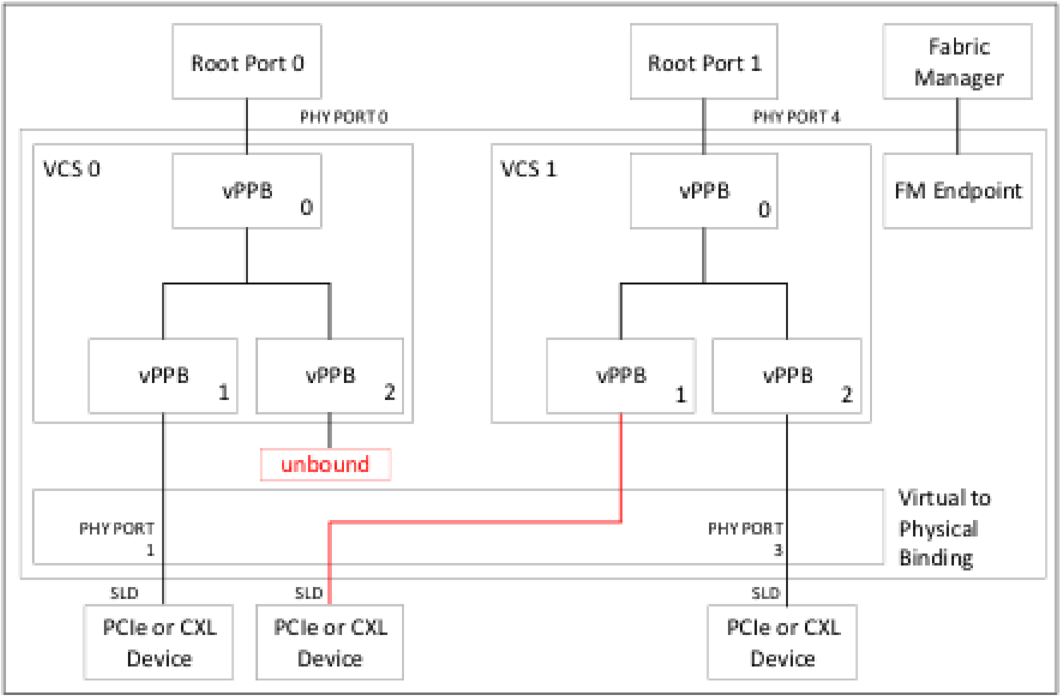


Figure 7-11 shows the state of the switch after the FM executes the bind command to connect VCS1.vPPB1 to the unbound physical port. The successful command execution results in the switch sending a hot-add indication to Host 1. Enumeration, configuration, and operation of the host and Type 3 device is identical to a hot-add of a device.

Figure 7-11. Example of CXL Switch Configuration after a Bind Command



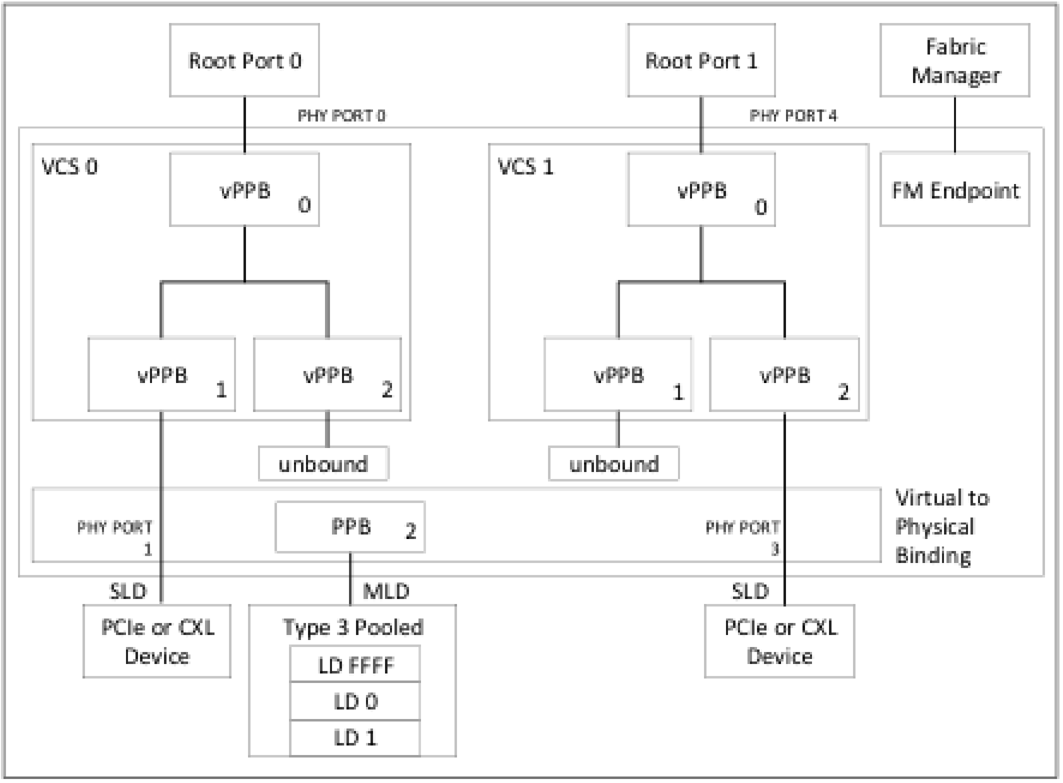
7.2.3.2 Binding and Unbinding of a Pooled Device

A pooled device contains multiple Logical Devices so that traffic over the physical port can be associated with multiple DS vPPBs. The switch behavior for binding and unbinding of an MLD component is similar to that of an SLD component, but with some notable differences:

1. The physical link cannot be impacted by binding and unbinding of a Logical Device within an MLD component. Thus, PERST#, Hot Reset, and Link Disable cannot be asserted, and there must be no impact to the traffic of other VCSs during the bind or unbind commands.
2. The physical PPB for an MLD port is always owned by the FM. The FM is responsible for port link control, AER, DPC, etc., and manages it using the FM API.
3. The FM may need to manage the pooled device to change memory allocations, enable the LD, etc.

Figure 7-12 shows a CXL switch after boot and before binding of any LDs within the pooled device. Note that the FM is not a PCIe Root Port and that the switch is responsible for enumerating the FMLD after any physical reset since the switch is responsible for proxying commands from FM to the device. The PPB of an MLD port is always owned by the FM since the FM is responsible for configuration and error handling of the physical port. After linkup the FM is notified that it is a Type 3 pooled device.

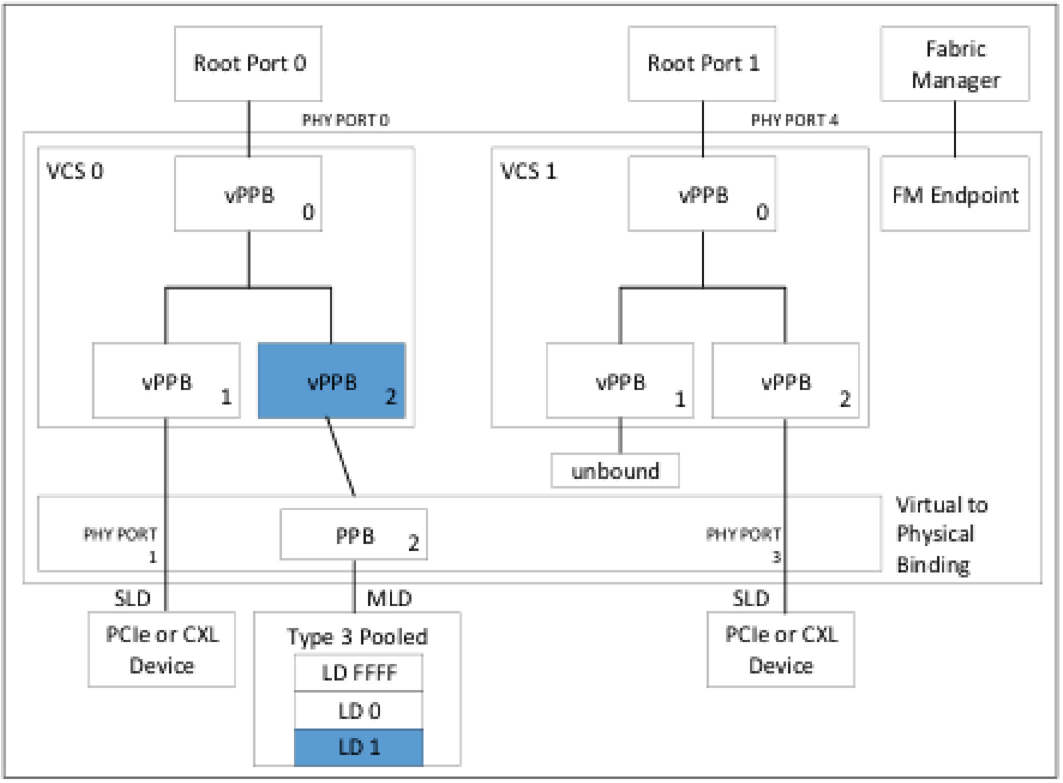
Figure 7-12. Example of a CXL Switch before Binding of LDs within Pooled Device



The FM configures the pooled device for Logical Device 1 (LD 1) and sets its memory allocation, etc. The FM performs a bind command for the unbound vPPB 2 in VCS 0 to LD 1 in the Type 3 pooled device. The switch performs the virtual-to-physical translations such that all CXL.io and CXL.mem transactions that target vPPB 2 in VCS 0 are routed to the MLD port with LD-ID set to 1. Additionally, all CXL.io and CXL.mem transactions from LD 1 in the pooled device are routed according to the host configuration of VCS 0. After binding, the vPPB notifies the VCS 0 host of a hot-add the same as if it were binding a vPPB to an SLD port.

Figure 7-13 shows the state of the switch after binding LD 1 to VCS 0.

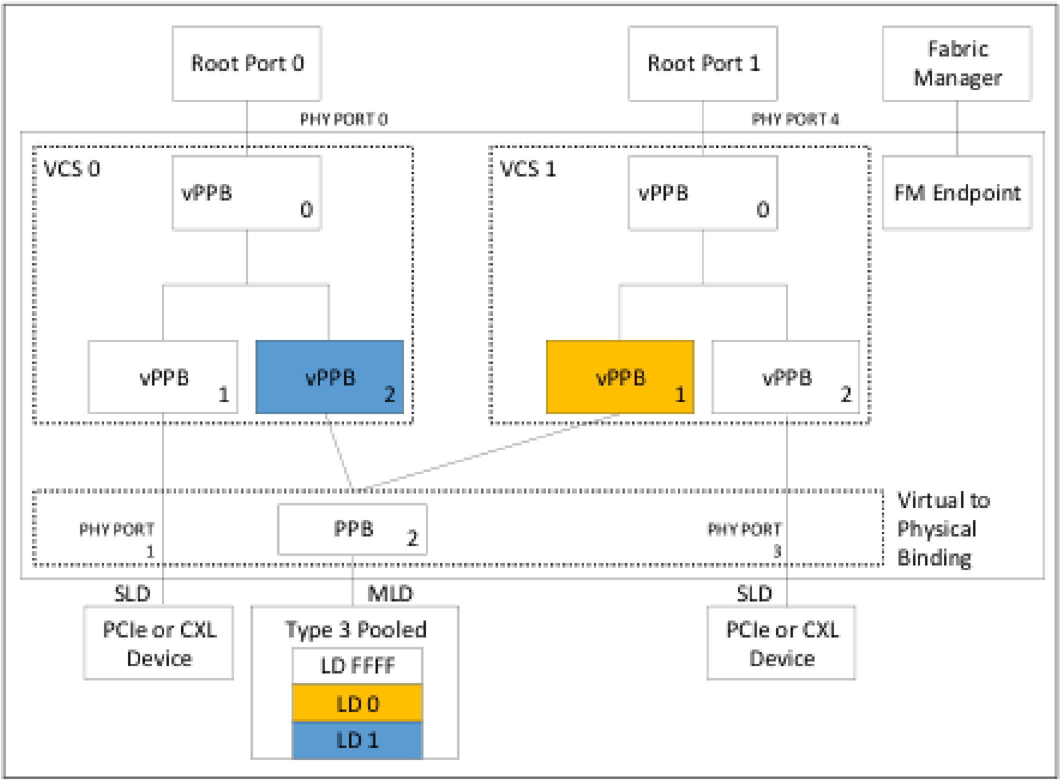
Figure 7-13. Example of a CXL Switch after Binding of LD-ID 1 within Pooled Device



The FM configures the pooled device for Logical Device 0 (LD 0) and sets its memory allocation, etc. The FM performs a bind command for the unbound vPPB 1 in VCS 1 to LD 0 in the Type 3 pooled device. The switch performs the virtual to physical translations such that all CXL.io and CXL.mem transactions targeting the vPPB in VCS 1 are routed to the MLD port with LD-ID set to 0. Additionally, all CXL.io and CXL.mem transactions from LD-ID = 0 in the pooled device are routed to the USP of VCS 1. After binding, the vPPB notifies the VCS 1 host of a hot-add the same as if it were binding a vPPB to an SLD port.

Figure 7-14 shows the state of the switch after binding LD 0 to VCS 1.

Figure 7-14. Example of a CXL Switch after Binding of LD-IDs 0 and 1 within Pooled Device



After binding LDs to vPPBs, the switch behavior is different from a bound SLD Port with respect to control, status, error notification, and error handling. Section 7.3.4 describes the differences in behavior for all bits within each register.

7.2.4 PPB and vPPB Behavior for MLD Ports

An MLD port provides a virtualized interface such that multiple vPPBs can access LDs over a shared physical interface. As a result, the characteristics and behavior of a vPPB that is bound to an MLD port are different than the behavior of a vPPB that is bound to an SLD port. This section defines the differences between them. If not mentioned in this section, the features and behavior of a vPPB that is bound to an MLD port are the same as those for a vPPB that is bound to an SLD port.

This section uses the following terminology:

- Hardwire to 0 refers to status and optional control register bits that are initialized to 0. Writes to these bits have no effect.
- The term 'Read/Write with no Effect' refers to control register bits where writes are recorded but have no effect on operation. Reads to those bits reflect the previously written value or the initialization value if it has not been changed since initialization.

7.2.4.1 MLD Type 1 Configuration Space Header

Table 7-2. MLD Type 1 Configuration Space Header

Register	Register Fields	FM-owned PPB	All Other vPPBs
Bridge Control Register	Parity Error Response Enable	Supported	Hardwire to 0s
	SERR# Enable	Supported	Read/Write with no effect
	ISA Enable	Not supported	Not supported
	Secondary Bus Reset (see Section 7.5 for SBR details for MLD ports)	Supported	Read/Write with no effect. Optional FM Event.

7.2.4.2 MLD PCIe-compatible Configuration Registers

Table 7-3. MLD PCIe-compatible Configuration Registers

Register/Capability Structure	Capability Register Fields	FM-owned PPB	All vPPBs Bound to the MLD Port
Command Register	I/O Space Enable	Hardwire to 0s	Hardwire to 0s
	Memory Space Enable	Supported	Supported per vPPB
	Bus Master Enable	Supported	Supported per vPPB
	Parity Error Response	Supported	Read/Write with no effect
	SERR# Enable	Supported	Supported per vPPB
	Interrupt Disable	Supported	Hardwire to 0s
Status Register	Interrupt Status	Hardwire to 0 (INTx is not supported)	Hardwire to 0s
	Master Data Parity Error	Supported	Hardwire to 0s
	Signaled System Error	Supported	Supported per vPPB
	Detected Parity Error	Supported	Hardwire to 0s

7.2.4.3 MLD PCIe Capability Structure

Table 7-4. MLD PCIe Capability Structure (Sheet 1 of 3)

Register/Capability Structure	Capability Register Fields	FM-owned PPB	All vPPBs Bound to the MLD Port
Device Capabilities Register	Max_Payload_Size Supported	Configured by the FM to the max value supported by switch hardware and min value configured in all vPPBs	Mirrors PPB
	Phantom Functions Supported	Hardwire to 0s	Hardwire to 0s
	Extended Tag Field Supported	Supported	Mirrors PPB
Device Control Register	Max_Payload_Size	Configured by the FM to Max_Payload Size Supported	Read/Write with no effect
Link Capabilities Register	Link Bandwidth Notification Capability	Hardwire to 0s	Hardwire to 0s

Table 7-4. MLD PCIe Capability Structure (Sheet 2 of 3)

Register/Capability Structure	Capability Register Fields	FM-owned PPB	All vPPBs Bound to the MLD Port
Link Capabilities	ASPM Support	No L0s support	No L0s support
	Clock Power Management	No PM L1 Substates support	No PM L1 Substates support
Link Control	ASPM Control	Supported	Switch only enables ASPM if all vPPBs that are bound to this MLD have enabled ASPM
	Link Disable	Supported	Switch handles it as an unbind by discarding all traffic to/from this LD-ID
	Retrain Link	Supported	Read/Write with no effect
	Common Clock Configuration	Supported	Read/Write with no effect
	Extended Synch	Supported	Read/Write with no effect
	Hardware Autonomous Width Disable	Supported	Read/Write with no effect
	Link Bandwidth Management Interrupt Enable	Supported	Read/Write with no effect
	Link Autonomous Bandwidth Interrupt Enable	Supported	Supported per vPPB. Each host can be notified of autonomous speed change
	DRS Signaling Control	Supported	Switch sends DRS after receiving DRS on the link and after binding of the vPPB to an LD
Link Status register	Current Link Speed	Supported	Mirrors PPB
	Negotiated Link Width	Supported	Mirrors PPB
	Link Training	Supported	Hardwire to 0s
	Slot Clock Configuration	Supported	Mirrors PPB
	Data Link Layer Active	Supported	Mirrors PPB
	Link Autonomous Bandwidth Status	Supported	Supported per vPPB
Slot Capabilities Register	Hot-Plug Surprise	Hardwire to 0s	Hardwired to 0s
	Physical Slot Number	Supported	Mirrors PPB
Slot Status Register	Attention Button Pressed	Supported	Mirrors PPB or is set by the switch on unbind
	Power Fault Detected	Supported	Mirrors PPB
	MRL Sensor Changed	Supported	Mirrors PPB
	Presence Detect Changed	Supported	Mirrors PPB or is set by the switch on unbind
	MRL Sensor State	Supported	Mirrors PPB
	Presence Detect State	Supported	Mirrors PPB or set by the switch on bind or unbind
	Electromechanical Interlock Status	Supported	Mirrors PPB
	Data Link Layer State Changed	Supported	Mirrors PPB or set by the switch on bind or unbind
Device Capabilities 2 Register	OBFF Supported	Hardwire to 0s	Hardwire to 0s

Table 7-4. MLD PCIe Capability Structure (Sheet 3 of 3)

Register/Capability Structure	Capability Register Fields	FM-owned PPB	All vPPBs Bound to the MLD Port
Device Control 2 Register	ARI Forwarding Enable	Supported	Supported per vPPB
	Atomic Op Egress Blocking	Supported	Mirrors PPB. Read/Write with no effect
	LTR Mechanism Enabled	Supported	Supported per vPPB
	Emergency Power Reduction Request	Supported	Read/Write with no effect. Optional FM notification.
	End-End TLP Prefix Blocking	Supported	Mirrors PPB. Read/Write with no effect
Link Control 2 Register	Target Link Speed	Supported	Read/Write with no effect. Optional FM notification.
	Enter Compliance	Supported	Read/Write with no effect
	Hardware Autonomous Speed Disable	Supported	Read/Write with no effect. Optional FM notification.
	Selectable De-emphasis	Supported	Read/Write with no effect
	Transmit Margin	Supported	Read/Write with no effect
	Enter Modified Compliance	Supported	Read/Write with no effect
	Compliance SOS	Supported	Read/Write with no effect
	Compliance Preset/De-emphasis	Supported	Read/Write with no effect
Link Status 2 Register	Current De-emphasis Level	Supported	Mirrors PPB
	Equalization 8.0 GT/s Complete	Supported	Mirrors PPB
	Equalization 8.0 GT/s Phase 1 Successful	Supported	Mirrors PPB
	Equalization 8.0 GT/s Phase 2 Successful	Supported	Mirrors PPB
	Equalization 8.0 GT/s Phase 3 Successful	Supported	Mirrors PPB
	Link Equalization Request 8.0 GT/s	Supported	Read/Write with no effect
	Retimer Presence Detected	Supported	Mirrors PPB
	Two Retimers Presence Detected	Supported	Mirrors PPB
	Crosslink Resolution	Hardwire to 0s	Hardwire to 0s
	Downstream Component Presence	Supported	Reflects the binding state of the vPPB
	DRS Message Received	Supported	Switch sends DRS after receiving DRS on the link and after binding of the vPPB to an LD

7.2.4.4 MLD PPB Secondary PCIe Capability Structure

All fields in the Secondary PCIe Capability Structure for a Virtual PPB shall behave identically to PCIe except the following:

Table 7-5. MLD Secondary PCIe Capability Structure

Register/Capability Structure	Capability Register Fields	FM-owned PPB	All vPPBs Bound to the MLD Port
Link Control 3 Register	Perform Equalization	Supported	Read/Write with no effect
	Link Equalization Request Interrupt Enable	Supported	Read/Write with no effect
	Enable Lower SKP OS Generation Vector	Supported	Read/Write with no effect
Lane Error Status Register	All fields	Supported	Mirrors PPB
Lane Equalization Control Register	All fields	Supported	Read/Write with no effect
Data Link Feature Capabilities Register	All fields	Supported	Hardwire to 0s
Data Link Feature Status Register	All fields	Supported	Hardwire to 0s

7.2.4.5 MLD Physical Layer 16.0 GT/s Extended Capability

All fields in the Physical Layer 16.0 GT/s Extended Capability Structure for a Virtual PPB shall behave identically to PCIe except the following:

Table 7-6. MLD Physical Layer 16.0 GT/s Extended Capability

Register/Capability Structure	Capability Register Fields	FM-owned PPB	All vPPBs Bound to the MLD Port
16.0 GT/s Status Register	All fields	Supported	Mirrors PPB
16.0 GT/s Local Data Parity Mismatch Status Register	Local Data Parity Mismatch Status Register	Supported	Mirrors PPB
16.0 GT/s First Retimer Data Parity Mismatch Status Register	First Retimer Data Parity Mismatch Status	Supported	Mirrors PPB
16.0 GT/s Second Retimer Data Parity Mismatch Status Register	Second Retimer Data Parity Mismatch Status	Supported	Mirrors PPB
16.0 GT/s Lane Equalization Control Register	Downstream Port 16.0 GT/s Transmitter Preset	Supported	Mirrors PPB

7.2.4.6 MLD Physical Layer 32.0 GT/s Extended Capability

Table 7-7. MLD Physical Layer 32.0 GT/s Extended Capability

Register/Capability Structure	Capability Register Fields	FM-owned PPB	All vPPBs Bound to the MLD Port
32.0 GT/s Capabilities Register	All fields	Supported	Mirrors PPB
32.0 GT/s Control Register	All fields	Supported	Read/Write with no effect
32.0 GT/s Status Register	Link Equalization Request 32.0 GT/s	Supported	Read/Write with no effect
	All fields except Link Equalization Request 32.0 GT/s	Supported	Mirrors PPB
Received Modified TS Data 1 Register	All fields	Supported	Mirrors PPB
Received Modified TS Data 2 Register	All fields	Supported	Mirrors PPB
Transmitted Modified TS Data 1 Register	All fields	Supported	Mirrors PPB
32.0 GT/s Lane Equalization Control Register	Downstream Port 32.0 GT/s Transmitter Preset	Supported	Mirrors PPB

7.2.4.7 MLD Lane Margining at the Receiver Extended Capability

Table 7-8. MLD Lane Margining at the Receiver Extended Capability

Register/Capability Structure	Capability Register Fields	FM-owned PPB	All vPPBs Bound to the MLD Port
Margining Port Status Register	All fields	Supported	Always indicates Margining Ready and Margining Software Ready
Margining Lane Control Register	All fields	Supported	Read/Write with no effect